Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/23/2025
PI Name:	Untaroiu, Costin Ph.D.		
Project Title:	Development of Injury Assessment Reference Values for the Lower Extremities Applicable to Spaceflight Use Cases		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	costin@vt.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	Virginia Polytechnic Institute & State University		
PI Address 1:	300 Turner Street NW, Ste 4200		
PI Address 2:			
PI Web Page:			
City:	Blacksburg	**State:**	VA
Zip Code:	24060	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/01/2026	**End Date:**	01/31/2028
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Period of performance (POP) has changed per NASA JSC. The new POP is 2/1/2026-1/31/28. The original POP was: 4/1/25-3/31/27 (Ed., 1/19/26).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC26K0269		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Injury prediction and prevention, in domains analogous to spaceflight dynamic events, commonly rely on the use of anthropomorphic test devices (ATD), also known as crash test dummies. While ATDs have been effective in reducing injury in several domains, their use is limited in spaceflight as the dynamic events often involve the use of a seat or spacesuit that may not be compatible with the ATDs, and the dynamic events themselves may not be reproducible in a laboratory setting. Finite element (FE) human body models, on the other hand, can be outfitted with a spacesuit and fit into any seat design that may be used. These models can then predict the loads, accelerations, etc., of the body during the dynamic phases of flight. However, the acceptable level of injury risk in spaceflight is much lower than in analogous domains, such as the automotive industry. There are currently no established thresholds for which loads on these human body models correspond to these lower levels of injury probability. This task seeks to identify these thresholds, also known as injury assessment reference values (IARVs), for the lower extremities. This process starts by first identifying spaceflight relevant datasets of injurious and non-injurious loading to the lower extremities. Next, the loading conditions from these datasets will be reconstructed with the human body models, and various outputs such as forces, bending moments, and/or stresses will be extracted. These outputs, also known as injury metrics, will then be correlated with the injury data to create injury risk curves. Finally, IARVs will be established corresponding to the NASA definition of acceptable risk. This will enable more precise prediction of lower extremity injury in a wider range of loading than is currently possible. This is a vital step in reducing injury risk during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The models used in this proposed work are a mechanism through which we can better understand the kinetics and kinematics of the human body during crewed vehicle landings or other dynamic events. These models allow predictive analytic biodynamic modeling, and they can allow for specific injury prediction by mission-phase, crew functionality post-landing, and vehicle design. The proposed work is a major step towards developing a standard methodology for using human body models to predict injury.
Task Progress:	New Project for FY2025
Bibliography Type:	Description: (Last Updated:)